FINAL PROJECT DOCUMENTATION

Project Title

Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets

Team ID : SWTID-2026-6274

Team Size : 4

Team Leader : DEBORA SHALINI T

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Team Member : MONISHA M S

Team Member : PRIYADHARSHINI S

Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets

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1. INTRODUCTION

1.1 Project Overview

In modern enterprise applications, maintaining accurate and consistent data is essential for efficient operations and decision-making. This project focuses on enforcing mandatory fields in forms using UI Policies to ensure that users provide all required information before submission.

Additionally, the project addresses the challenge of migrating configurations and customizations across different environments (development, testing, and production) using Update Sets. This ensures a structured and error-free deployment process.

The combination of UI Policies and Update Sets provides a robust solution for improving data quality and maintaining system consistency.

1.2 Purpose

The main purpose of this project is to:

- Ensure all required fields are filled before form submission
- Improve data accuracy and reliability
- Reduce manual errors during data entry
- Simplify the migration of system changes
- Maintain consistency across multiple environments

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2. IDEATION PHASE

2.1 Problem Statement

Organizations often face issues with incomplete or incorrect data due to users skipping important fields. This leads to:

- Poor data quality
- Errors in reports and workflows
- Increased manual correction efforts

Additionally, migrating changes manually between environments can:

- Cause inconsistencies
- Lead to deployment errors
- Increase development time

2.2 Empathy Map Canvas

Users Think: The system should be simple and guide them clearly
Users Feel: Frustrated when errors occur after submission
Users Say: “I want a system that prevents mistakes while filling forms” **Users Do:** Enter data quickly, sometimes skipping fields

This highlights the need for real-time validation and user-friendly enforcement mechanisms.

2.3 Brainstorming

Different approaches were considered:

- Client-side scripting
- Server-side validation
- UI Policies
- Data Policies

After evaluation, UI Policies were chosen because:

- They provide dynamic control on the user interface
- They are easy to configure without heavy coding
- They improve user experience

Update Sets were selected for migration because:

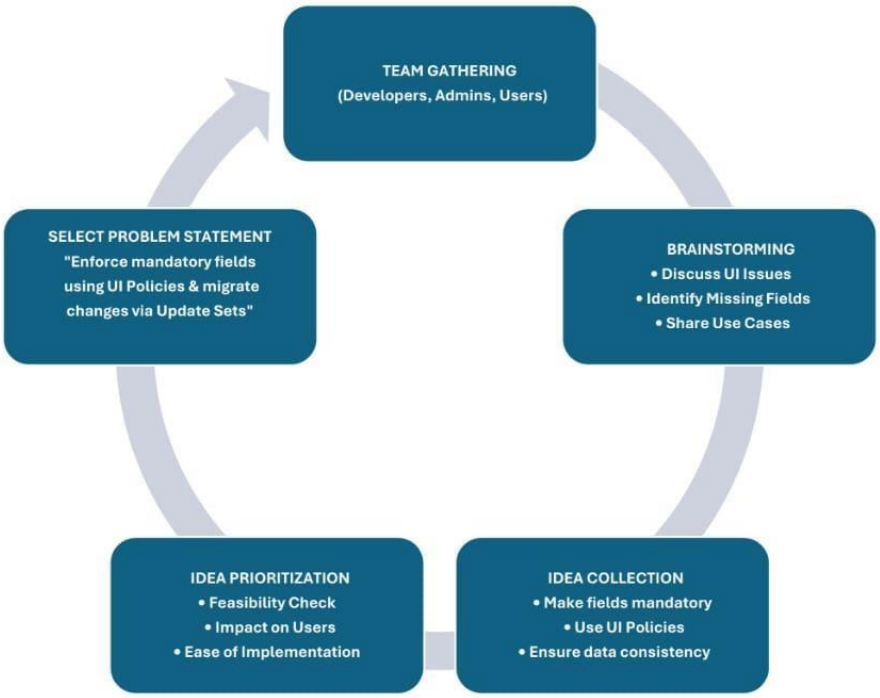
- They track configuration changes
- They allow smooth transfer between instances
- They reduce deployment risks

Ideation Phase

Brainstorm & Idea Prioritization

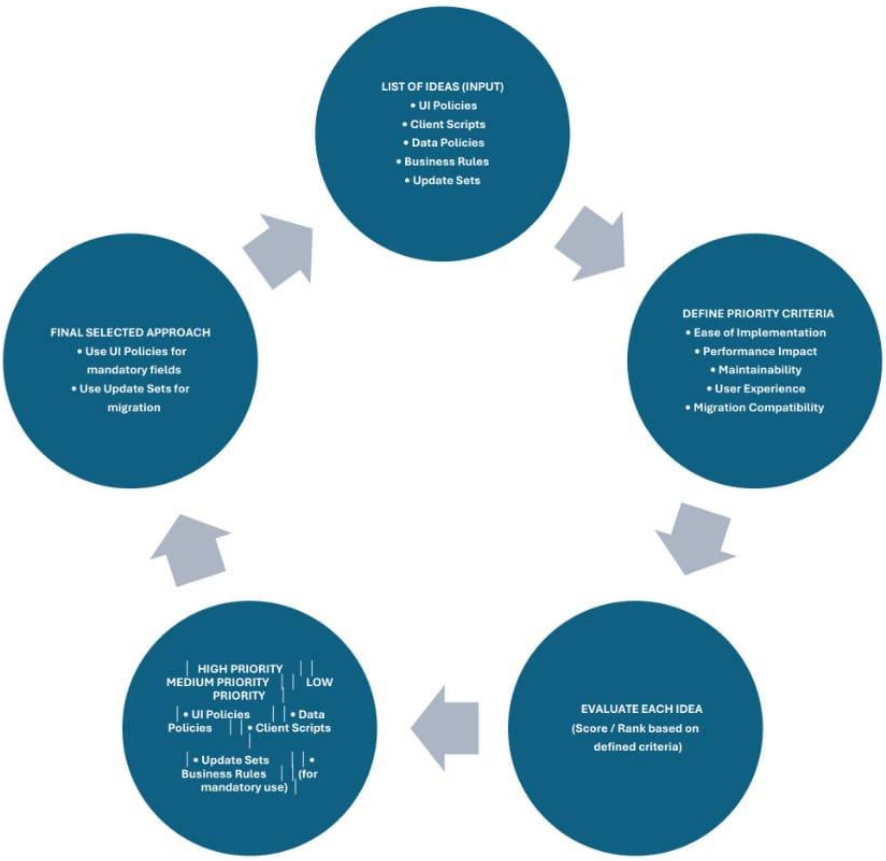
Brainstorm & Idea Prioritization is a two-part process used to generate ideas and then decide which ones are most valuable or worth pursuing.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

Step-3: Idea Prioritization



Ideation phase

Define the Problem Statements

Customer Problem Statement

Customer Problem Support refers to the process of helping customers identify, address, and resolve issues they face while using a product or service, ensuring a positive experience and satisfaction.

CURRENT SITUATION

- **Important fields left empty**
- **Inconsistent data entry**
- **Manual validation required**

▼

CUSTOMER PAIN

- **Poor data quality**
- **Process delays**
- **Reporting inaccuracies**
- **Increased rework**

▼

ROOT CAUSE

- No enforced mandatory fields
- Lack of UI-level validation
- No structured migration

▼

PROBLEM STATEMENT

Users are able to submit forms without filling required fields, leading to inconsistent and unreliable data across the system.

▼

BUSINESS IMPACT

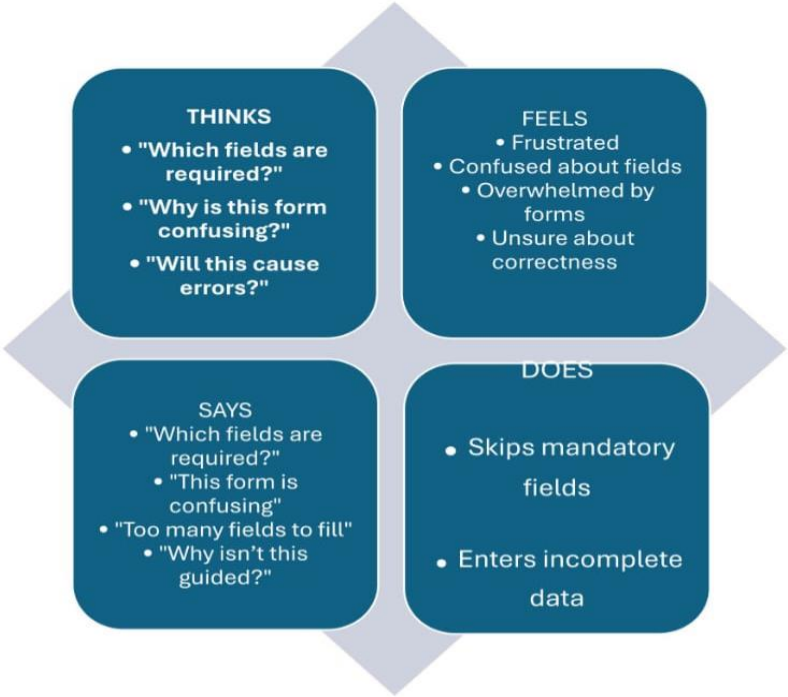
- Poor decision-making
- Compliance risks
- Inefficient workflows
- Difficulty in scaling

Ideation Phase

Empathize & Discover

EMPATHY MAP

An Empathy Map is a visual tool used to understand a customer or user by capturing what they think, feel, say, and do, along with their needs and pain points. It helps teams better understand user experiences and design solutions that truly meet their needs.



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User logs into the system
2. Opens a form
3. Starts entering data
4. UI Policies enforce mandatory fields dynamically
5. User fills all required fields
6. Form is submitted successfully
7. Data is stored in the system

3.2 Solution Requirement

The system must:

- Dynamically enforce mandatory fields
- Provide visual indicators for required fields
- Prevent submission if mandatory fields are empty
- Support easy configuration of UI Policies
- Enable migration using Update Sets
- Maintain system performance

3.3 Data Flow Diagram

Flow:
User Input → UI Policy Validation → Application Processing → Database Storage → Output Display

- UI Policies validate inputs at the front-end
- Only valid data is passed to the backend
- Data is securely stored and retrieved

3.4 Technology Stack

- **ServiceNow Platform**
- UI Policies (for front-end validation)

REQUIREMENT ANALYSIS PHASE-CUSTOMER JOURNEY MAP

	User opens form/task	Views form interface	Enters data in fields	Submits form	Data used in system
FEELINGS	Curious	Confused	Unsure	Dissatisfied
PAIN POINTS	No guidance	Unclear req.	Skips fields	Bad data
OPPORTUNITY	Guide users	Highlight mandatory	Use UI Policies	Enforce validation

USER STORIES

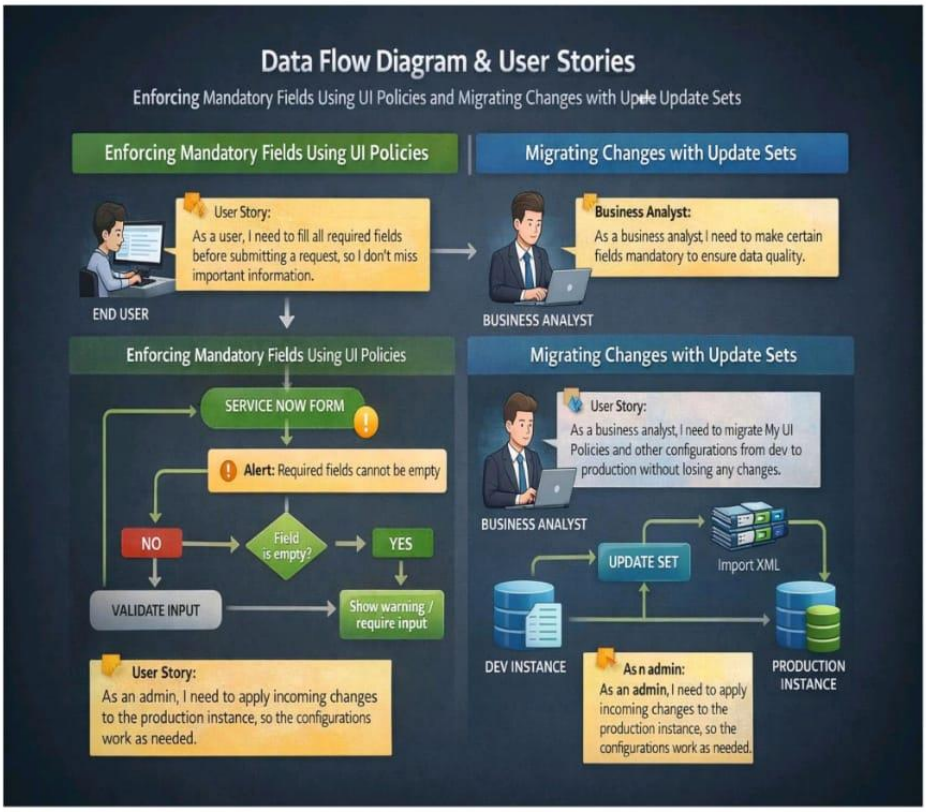
User Stories – Enforcing Mandatory Fields & Update Sets

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (End User)	Form Validation	USN-1	As a user, I want mandatory fields to be clearly marked so that I know what information is required.	Mandatory fields are highlighted (e.g., red asterisk)	High	Sprint-1
Customer (End User)	Form Validation	USN-2	As a user, I want to receive an alert if I leave mandatory fields empty so that I can correct them before submission.	Error message shown when required fields are empty	High	Sprint-1
Customer (End User)	Form Submission	USN-3	As a user, I want to submit the form only after all required fields are filled so that my request is successfully processed.	Form submission blocked until all required fields are completed	High	Sprint-1
Customer (End User)	User Guidance	USN-4	As a user, I want guidance messages on fields so that I understand what data to enter.	Tooltips/help text available for fields	Medium	Sprint-2
Business Analyst	UI Policy Configuration	USN-5	As a business analyst, I want to configure UI Policies to make fields mandatory so that data quality is ensured.	UI Policies enforce mandatory fields dynamically	High	Sprint-1
Business Analyst	UI Policy Testing	USN-6	As a business analyst, I want to test UI Policies so that they work correctly before deployment.	Policies tested successfully in dev environment	High	Sprint-1
Administrator	Update Set Creation	USN-7	As an admin, I want to capture UI Policy changes in an Update Set so that configurations can be migrated.	Changes recorded in update set	High	Sprint-1
Administrator	Migration	USN-8	As an admin, I want to migrate update sets from dev to production so that configurations are ap-plled in live system.	Update set successfully moved and commited	High	Sprint-2
Customer Care Executive	Validation	USN-10	As a support user, I want complete and accurate form data so that I can process requests effclen-ity.	All submitted records have required fields filled	Medium	Sprint-2
Customer (Web User)	Form Experience	USN-11	As a web user, I want a smooth form experience with validation so that I can submit requests without confusion.	Form validation works consistently across devices	Medium	Sprint-2

Data Flow Diagram & User Stories

DATA FLOW DAIGRAMS

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

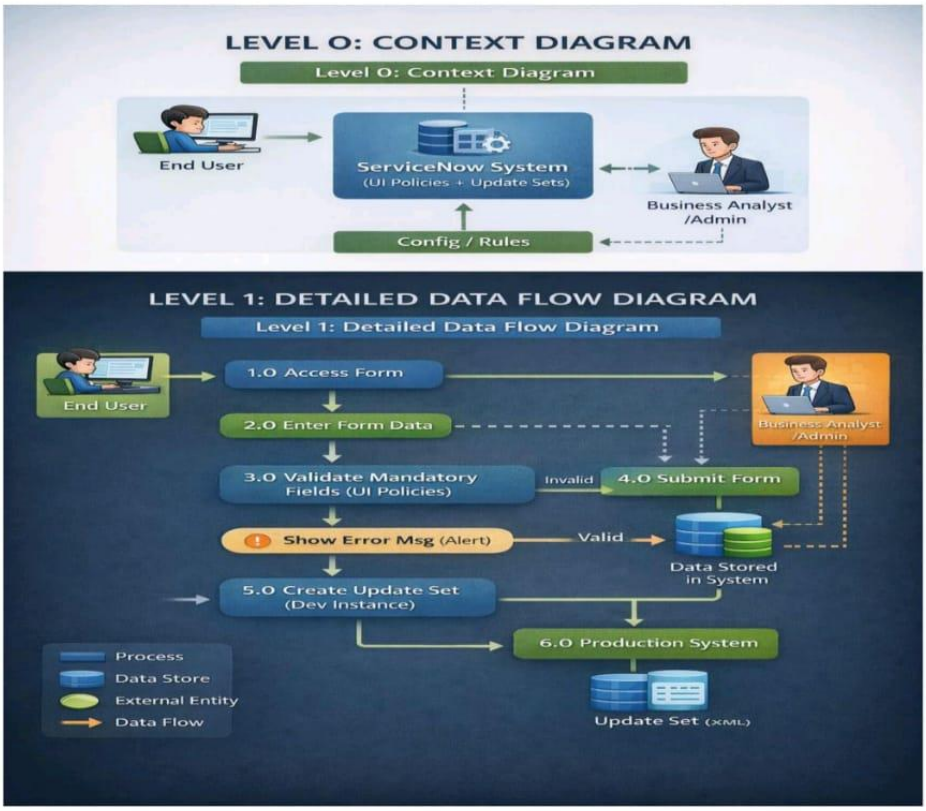


🔹 LEVEL 0 – Context Diagram

This shows the system as a whole with external entities.

🔹 LEVEL 1 – Detailed Data Flow Diagram

Breaks the system into internal processes.



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Solution Requirements (Functional & Non-functional)

Functional Requirements:

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Enforcing Mandatory Fields (UI Policies)	• Identify required fields in forms • Configure UI Policies to make fields mandatory • Display visual indicators (e.g., red asterisk) • Show error messages when fields are empty
FR-2	Form Validation	• Prevent form submission if mandatory fields are not filled • Validate fields dynamically on change • Provide real-time user feedback
FR-3	UI Policy Testing	• Test UI Policies in development eniironment • Ensure policies trigger correctly under conditions
FR-4	Update Set Creation	• Capture UI Policy and configuration changes in Update Sets • Maintain version control of changes
FR-5	Migration via Update Sets	• Export Update Sets from development instance • Import Update Sets into test/production instance • Commit Update Sets after validation
FR-6	Update Set Validation	• Preview Update Sets for conflicts/errors • Resolve issues before deployment
FR-7	Data Integrity	• Ensure all submitted records contain mandatory data
FR-7	Data Integrity	• Maintain consistency across environments

Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	UI policies are easy to use and clearly show mandatory fields.
NFR-2	**Security**	Only authorized users can modify UI policies and update sets.
NFR-3	**Reliability**	Mandatory fields are always enforced and update sets work correctly.
NFR-4	**Performance**	UI policies should not slow down the system.
NFR-5	**Availability**	System should always be available for validation and migration.
NFR-6	**Availability**	System should always be available for validation and migration.
NFR-6	**Scalability**	Supports more UI policies and large update sets as system grows.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem of incomplete data is effectively solved using UI Policies, which enforce rules at the user interface level. Update Sets address the challenge of migrating configurations safely and efficiently.

4.2 Proposed Solution

- Create UI Policies to make specific fields mandatory based on conditions
- Configure actions to display error messages or highlight fields
- Use Update Sets to capture all changes
- Move Update Sets between environments for deployment

4.3 Solution Architecture

Layers:

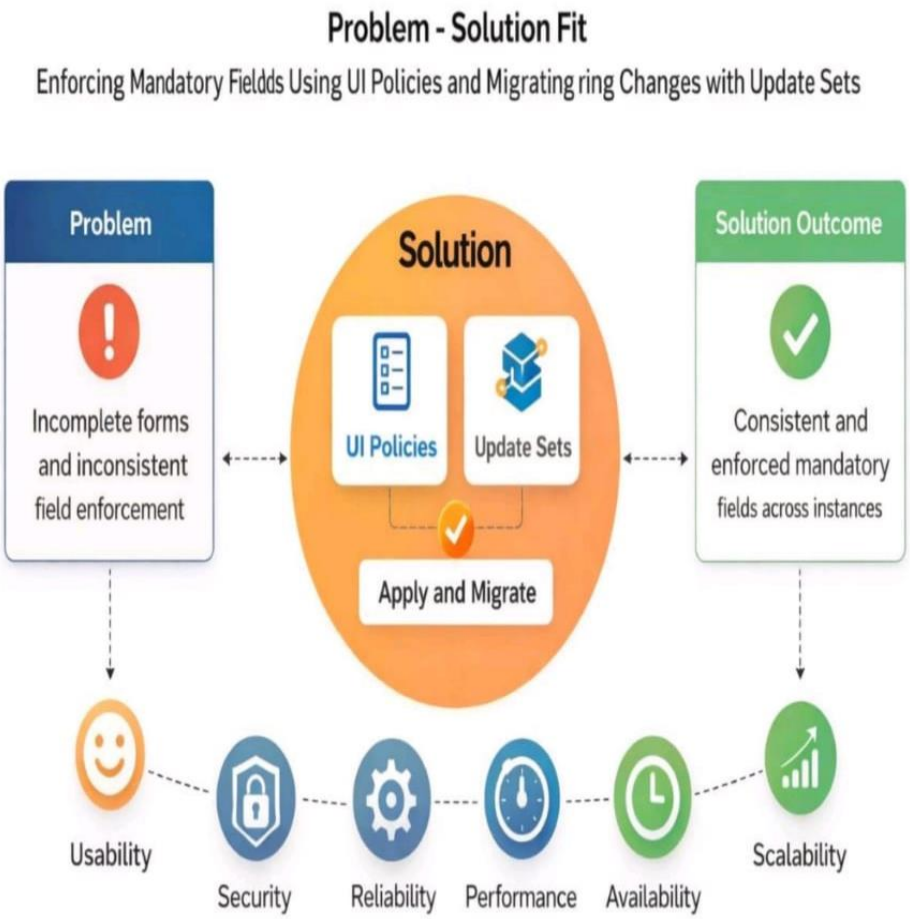
1. User Interface Layer – Forms and fields
2. UI Policy Layer – Validation rules
3. Application Layer – Business logic
4. Database Layer – Data storage
5. Migration Layer – Update Sets

4.4 Purpose:

❑ Solve complex problems in a way that fits the state of your customers.

❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.

❑ Sharpen your communication and marketing strategy with the right triggers and messaging.



Project Design Phase

Proposed Solution

Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets

S.No.	Parameter	Description
1	**Problem Statement** (Problem to be solved)	In many systems, users may submit forms without filling all required fields, leading to incomplete or inconsistent data. Additionally, changes made in one environment (like development) are difficult to move accurately to another (like production) without a proper migration method.
	Idea / Solution Description	Use UI Policies to make specific fields mandatory based on conditions, ensuring users can provide all required information before submission. Use Update Sets to capture configuration changes and migrate them safely from one instance to another (like production).
	Novelty / Uniqueness	Combines dynamic form validaton (UI Policies) with structured change management (Update Sets). This ensures both data quality and smooth deployment, reducing manual errors and improving system reliability.
	Social Impact / Customer Satisfaction	Improves user experience by guiding users to enter correct data. Reduces errors, increases efficiency, and ensures reliable information, leading to better customer satisfaction and trust in the system.
	Business Model (Revenue Model)	Helps organizations save costs by reducing data errors and rework. Improves productivity and system efficiency, which indirectly increases revenue. Can be part of enterprise IT service management solutions.
	Scalability of the Solution	Highly scalable as UI Policies can be applied across multiple forms and modules, and Update Sets support easy migration across multiple environments. Suitable for both small and large organizations.

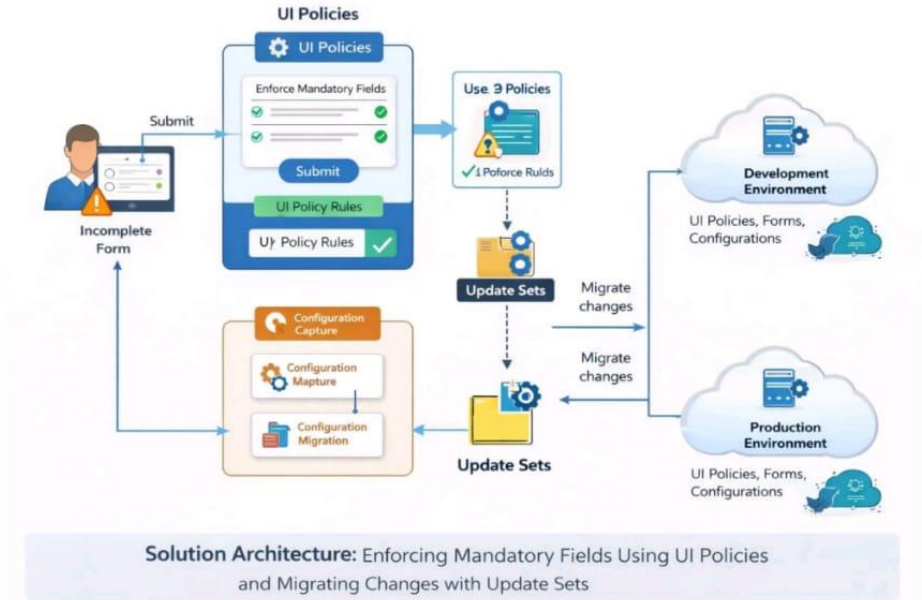
Project Design Phase

Solution Architecture

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions.

Example - Solution Architecture Diagram:



Solution Architecture: Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed in phases:

1. Requirement analysis
2. Design of UI Policies
3. Implementation
4. Testing
5. Deployment using Update Sets

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

- Verified mandatory fields are enforced
- Checked error messages and field highlighting
- Ensured form submission is blocked when fields are empty

6.2 Performance Testing

- Tested system response time
- Ensured UI Policies do not slow down form loading
- Verified smooth functioning under multiple users

Performance Testing

Enforcing Mandatory Fields Using UI Policies & Migrating

Changes with Update Sets

1. Objective

• Ensure UI Policies and form validations perform

efficiently

• Verify that mandatory field enforcement does not slow down the system

• Validate smooth migration using Update Sets

2. Key Performance Areas

Form Load Performance

• Measure time taken to load forms with UI Policies applied

• Ensure no delay due to multiple policies

Expected Result: Form loads within acceptable time (e.g.,

< 2–3 seconds)

UI Policy Execution Time

• Check how quickly UI Policies trigger when fields change

• Validate real-time response

Expected Result: Immediate execution without lag

Form Submission Time

• Measure time taken to submit form after validation

• Ensure mandatory checks do not delay submission Expected Result: Submission completes quickly without errors

Concurrent Users Load

• Test system with multiple users filling forms simultaneously

Expected Result: No performance degradation under load

Update Set Migration Performance

• Measure time taken to move Update Sets between instances

• Check for delays during preview and commit

Expected Result: Smooth and timely migration

User Acceptance Testing (UAT)

Enforcing Mandatory Fields Using UI Policies & Migrating Changes with Update Sets

1. Objective

• Ensure the solution meets business requirements

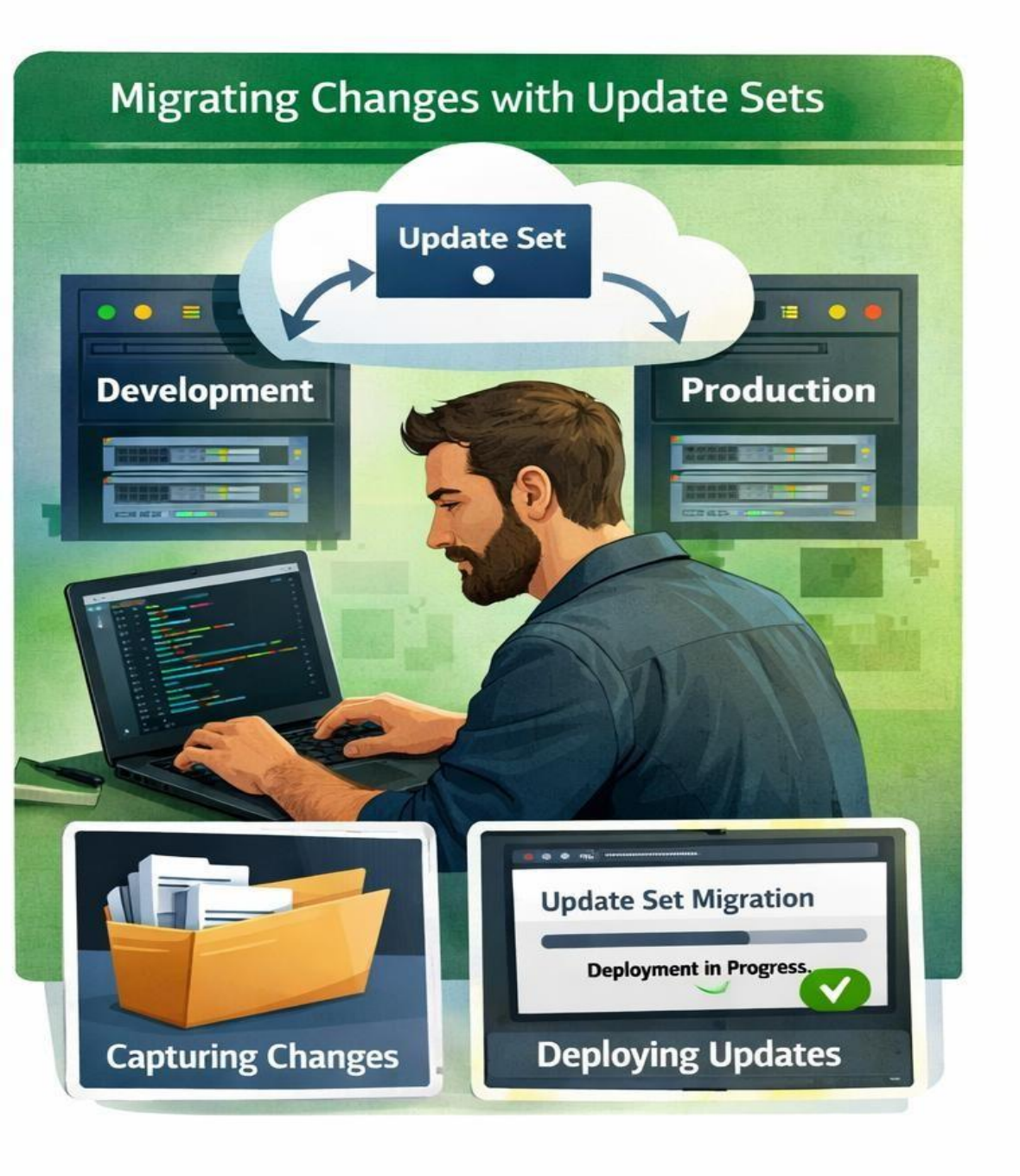
• Validate that mandatory fields are enforced correctly

• Confirm Update Sets migration works as expected

• Get final approval from business users before production

2. UAT Scope

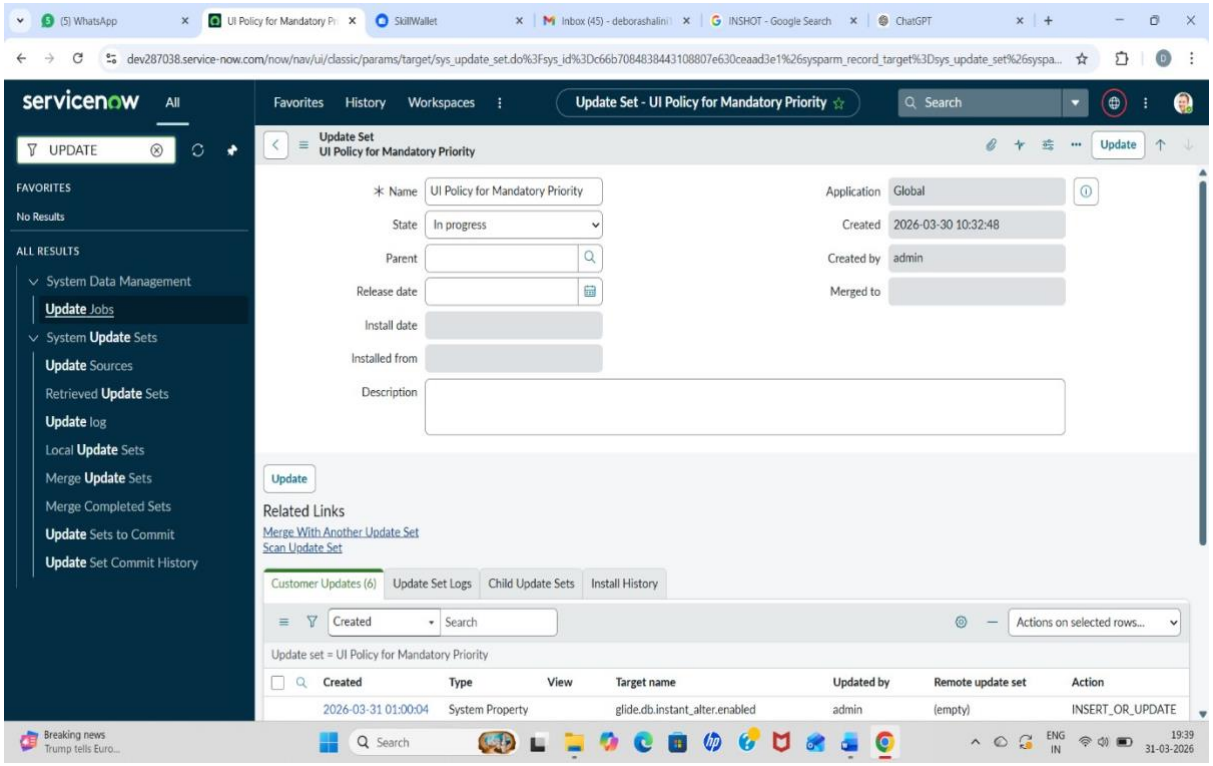
- UI Policy behavior on forms
- Mandatory field validation
- Error messages & user guidance
- Form submission process
- Update Set migration (Dev → Test → Prod)

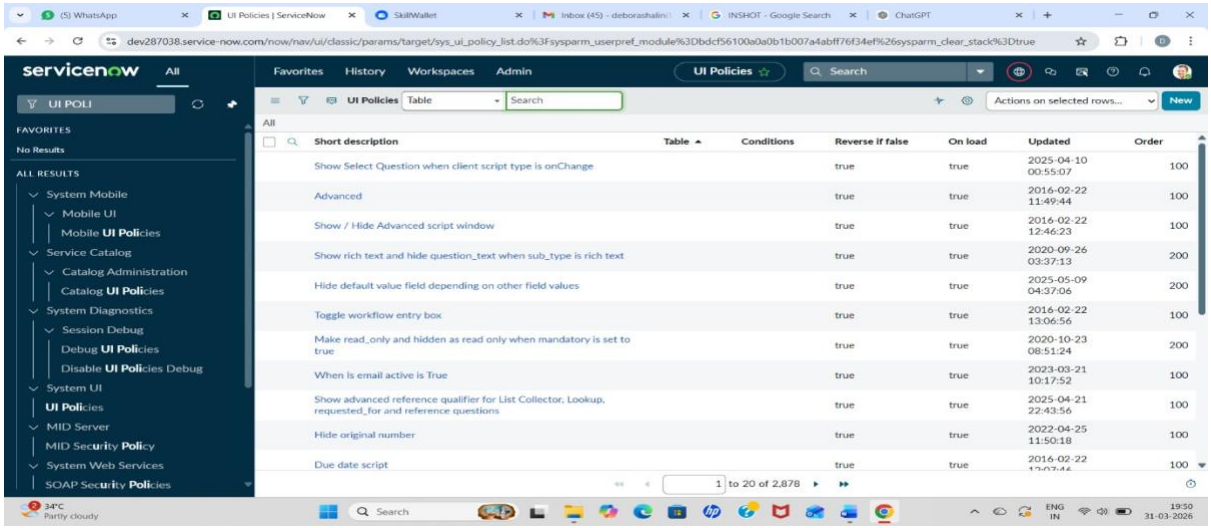

Migrating Changes with Update Sets
Update Set
Development
Production
Capturing Changes
Update Set Migration
Deployment in Progress.
Deploying Updates

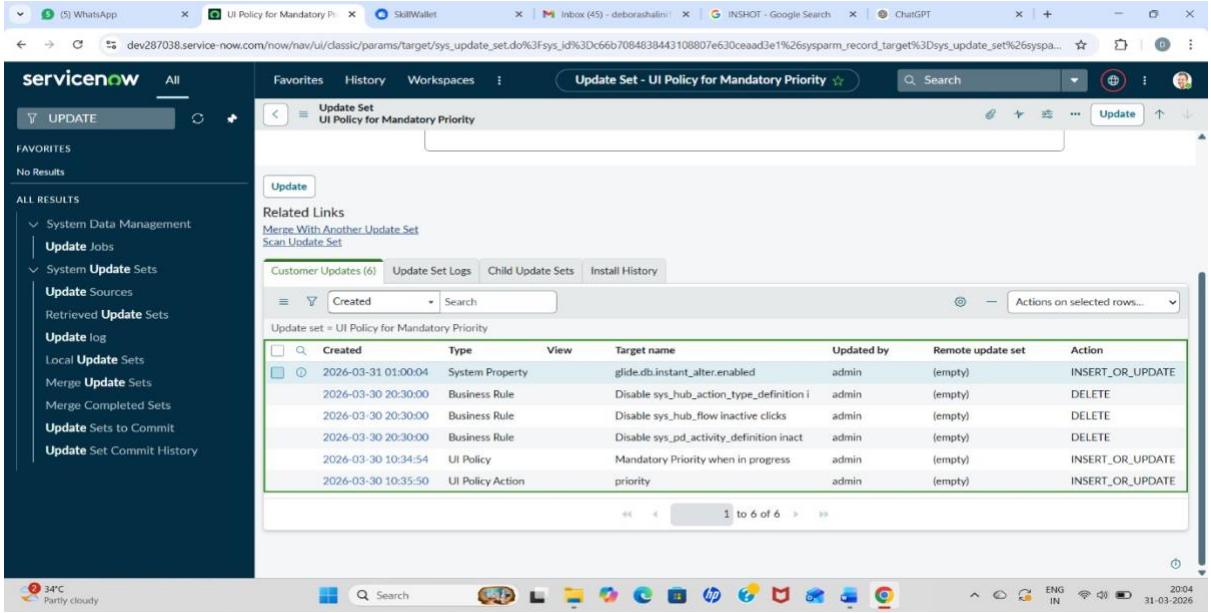
7. RESULTS

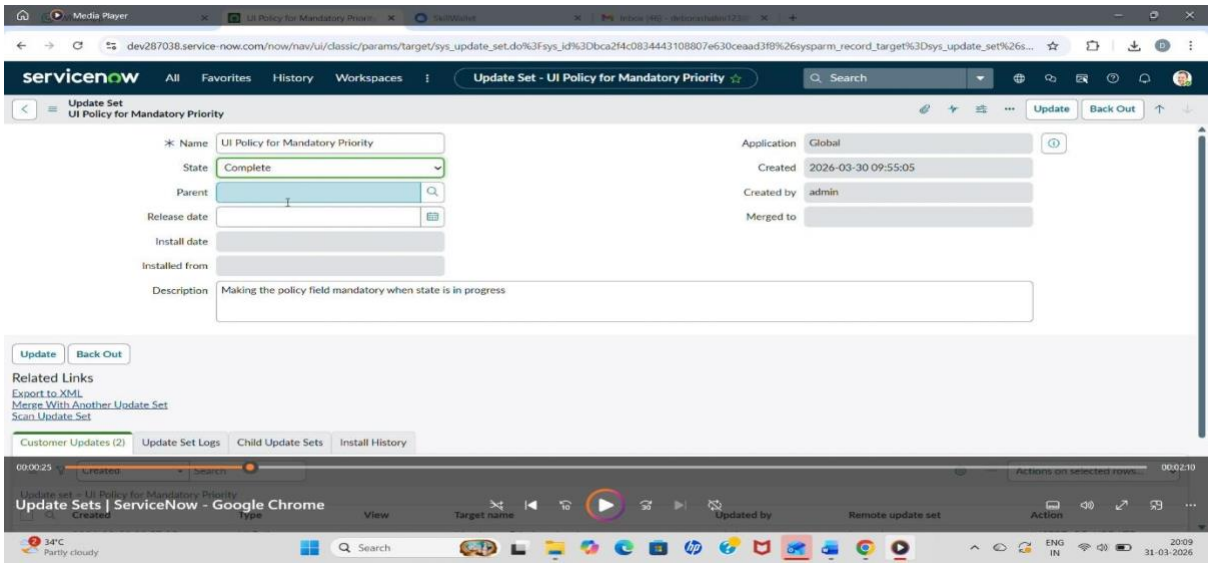
7.1 Output Screenshots

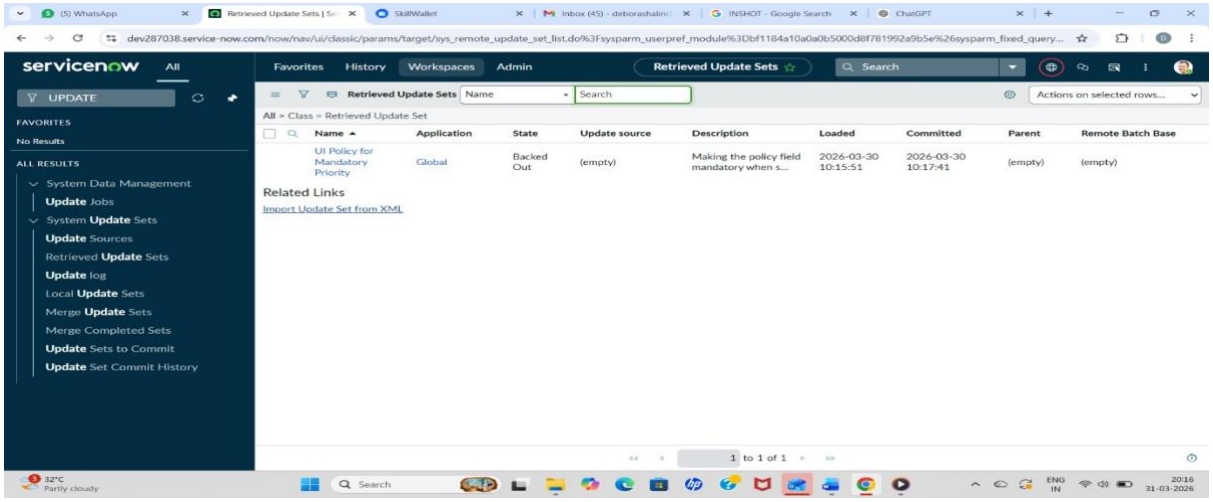
- Forms with mandatory fields highlighted
- Error messages for missing inputs
- Successful submission after validation
- Update Set migration confirmation

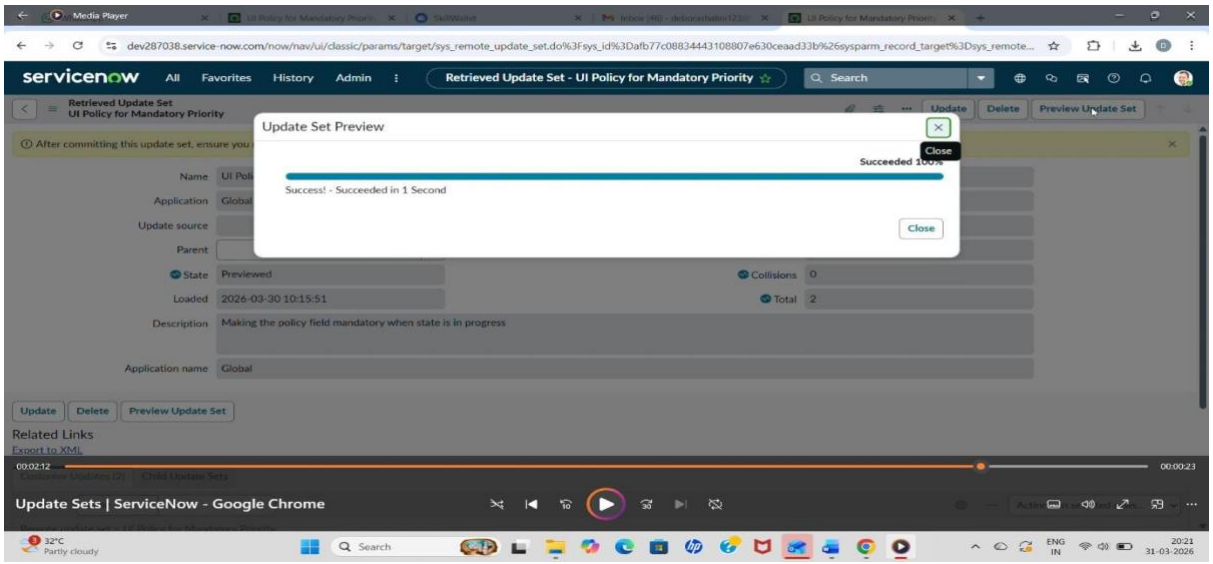


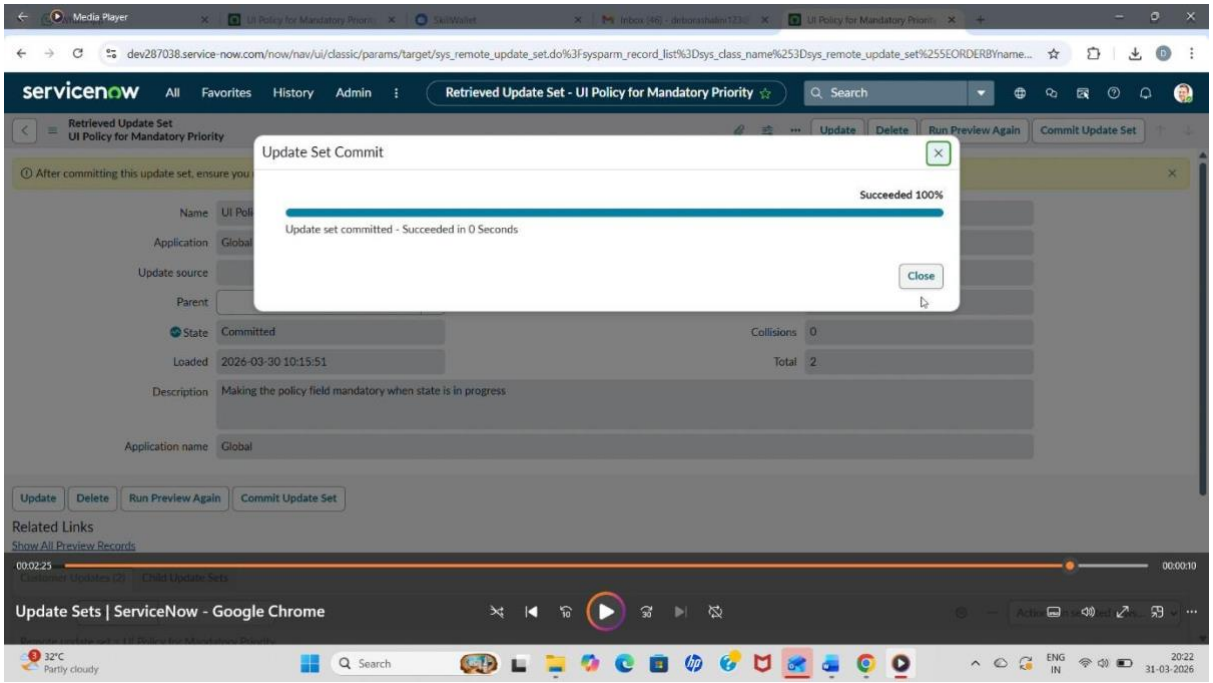

servicenow
All
Favorites
History
Workspaces
Admin
UI Policies
Search
UI POLI
FAVORITES
No Results
ALL RESULTS
System Mobile
Mobile UI
Mobile UI Policies
Service Catalog
Catalog Administration
Catalog UI Policies
System Diagnostics
Session Debug
Debug UI Policies
Disable UI Policies Debug
System UI
UI Policies
MID Server
MID Security Policy
System Web Services
SOAP Security Policies
UI Policies Table
Actions on selected rows...
New
Short description
Table
Conditions
Reverse if false
On load
Updated
Order
Show Select Question when client script type is onChange
Advanced
Show / Hide Advanced script window
Show rich text and hide question_text when sub_type is rich text
Hide default value field depending on other field values
Toggle workflow entry box
Make read_only and hidden as read only when mandatory is set to true
When is email active is True
Show advanced reference qualifier for List Collector, Lookup, requested_for and reference questions
Hide original number
Due date script
1 to 20 of 2,878


servicenow
All
Favorites
History
Workspaces
Update Set - UI Policy for Mandatory Priority
Search
UPDATE
FAVORITES
No Results
ALL RESULTS
System Data Management
Update Jobs
System Update Sets
Update Sources
Retrieved Update Sets
Update log
Local Update Sets
Merge Update Sets
Merge Completed Sets
Update Sets to Commit
Update Set Commit History
Update Set
UI Policy for Mandatory Priority
Update
Related Links
Merge With Another Update Set
Scan Update Set
Customer Updates (6)
Update Set Logs
Child Update Sets
Install History
Created
Actions on selected rows...
Update set = UI Policy for Mandatory Priority
Type
View
Target name
Updated by
Remote update set
Action
2026-03-31 01:00:04
System Property
glide.db.instant_alter.enabled
admin
(empty)
INSERT_OR_UPDATE
2026-03-30 20:30:00
Business Rule
Disable sys_hub_action_type_definition i
admin
(empty)
DELETE
2026-03-30 20:30:00
Business Rule
Disable sys_hub_flow inactive clicks
admin
(empty)
DELETE
2026-03-30 20:30:00
Business Rule
Disable sys_pd_activity_definition inact
admin
(empty)
DELETE
2026-03-30 10:34:54
UI Policy
Mandatory Priority when in progress
admin
(empty)
INSERT_OR_UPDATE
2026-03-30 10:35:50
UI Policy Action
priority
admin
(empty)
INSERT_OR_UPDATE
1 to 6 of 6


Update Set - UI Policy for Mandatory Priority
Update Set
UI Policy for Mandatory Priority
Name
UI Policy for Mandatory Priority
State
Complete
Parent
Release date
Install date
Installed from
Description
Making the policy field mandatory when state is in progress
Application
Global
Created
2026-03-30 09:55:05
Created by
admin
Merged to
Update
Back Out
Related Links
Export to XML
Merge With Another Update Set
Scan Update Set
Customer Updates (2)
Update Set Logs
Child Update Sets
Install History
Update Sets | ServiceNow - Google Chrome


Retrieved Update Sets
Name
UI Policy for Mandatory Priority
Application
Global
State
Backed Out
Update source
(empty)
Description
Making the policy field mandatory when s...
Loaded
2026-03-30 10:15:51
Committed
2026-03-30 10:17:41
Parent
(empty)
Remote Batch Base
(empty)
Related Links
Import Update Set from XML


Retrieved Update Set - UI Policy for Mandatory Priority
Update Set Preview
Succeeded 100%
Success! - Succeeded in 1 Second
Close
Name
Application
Update source
Parent
State
Previewed
Collisions
0
Loaded
2026-03-30 10:15:51
Total
2
Description
Making the policy field mandatory when state is in progress
Application name
Global
Update
Delete
Preview Update Set
Related Links
Export to XML
Update Sets | ServiceNow - Google Chrome


Retrieved Update Set - UI Policy for Mandatory Priority
Update Set Commit
Succeeded 100%
Update set committed - Succeeded in 0 Seconds
Close
Name
Application
Update source
Parent
State
Committed
Collisions
0
Loaded
2026-03-30 10:15:51
Total
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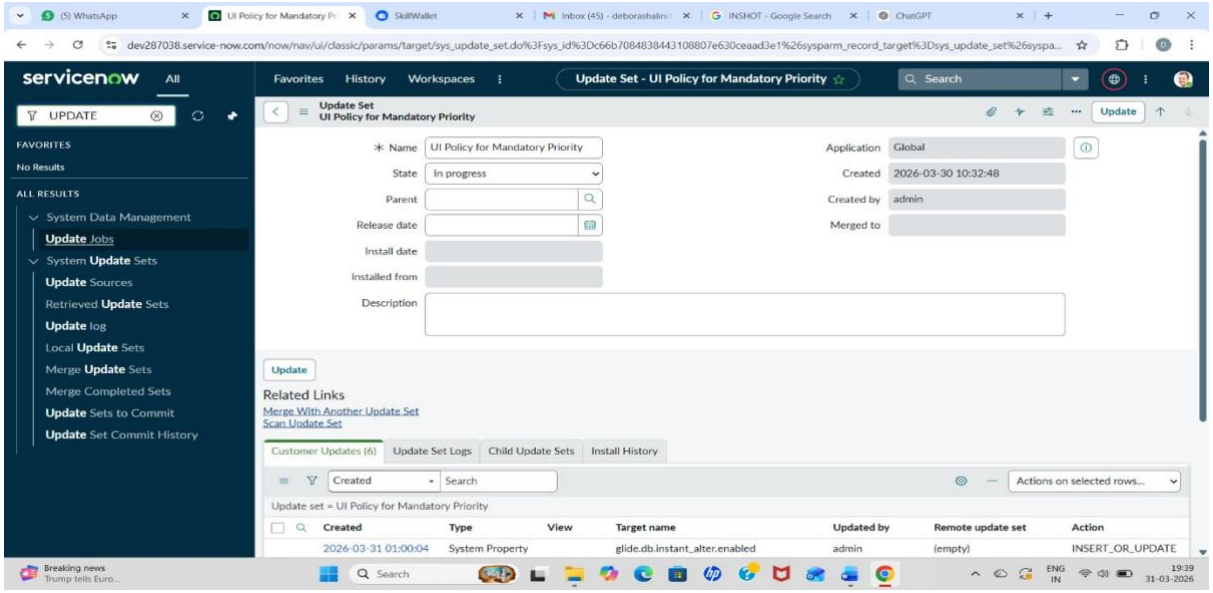
Description
Making the policy field mandatory when state is in progress
Application name
Global
Update
Delete
Run Preview Again
Commit Update Set
Related Links
Show All Preview Records
Update Sets | ServiceNow - Google Chrome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Update Set - UI Policy for Mandatory Priority
Update Set
UI Policy for Mandatory Priority
Name
UI Policy for Mandatory Priority
State
In progress
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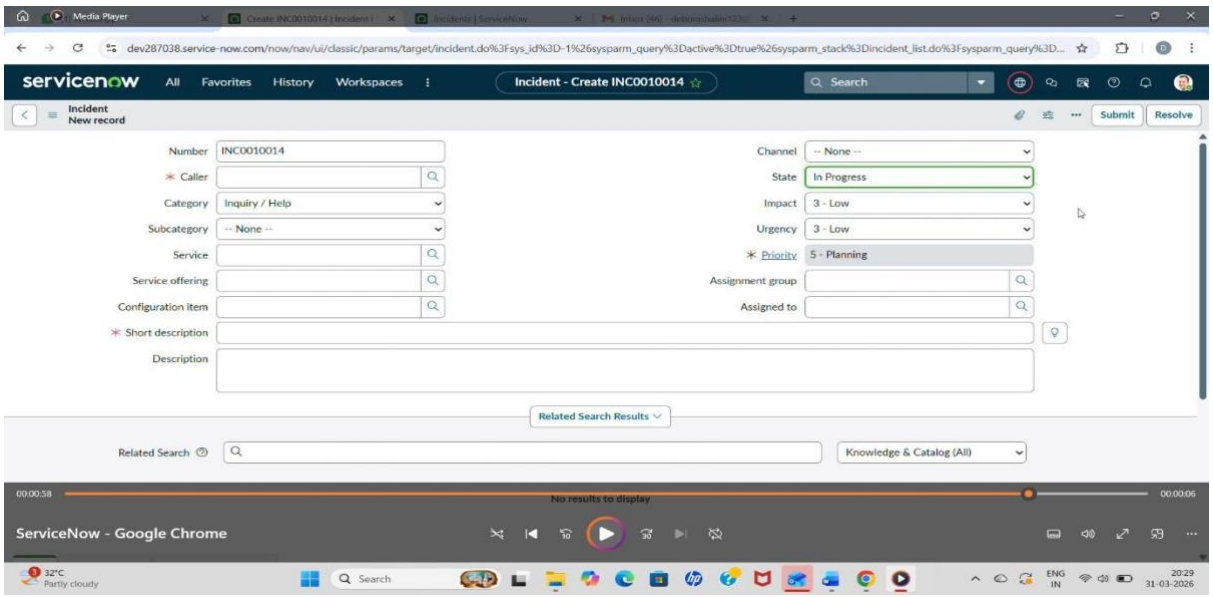
Release date
Install date
Installed from
Description
Application
Global
Created
2026-03-30 10:32:48
Created by
admin
Merged to
Update
Related Links
Merge With Another Update Set
Scan Update Set
Customer Updates (6)
Update Set Logs
Child Update Sets
Install History
Update set = UI Policy for Mandatory Priority
Created
Type
View
Target name
Updated by
Remote update set
Action
2026-03-31 01:00:04
System Property
glide.db.instant_alter.enabled
admin
(empty)
INSERT_OR_UPDATE


servicenow
Favorites
History
Workspaces
Incident - Create INC0010014
Incident
New record
Submit
Resolve
Number
INC0010014
Caller
Category
Inquiry / Help
Subcategory
-- None --
Service
Service offering
Configuration item
Short description
Description
Channel
-- None --
State
In Progress
Impact
3 - Low
Urgency
3 - Low
Priority
5 - Planning
Assignment group
Assigned to
Related Search Results
Related Search
Knowledge & Catalog (All)
No results to display
ServiceNow - Google Chrome

8. ADVANTAGES & DISADVANTAGES

Advantages

- Improved data accuracy
- Reduced manual errors
- Easy implementation
- Better user experience
- Efficient migration process

Disadvantages

- Requires proper configuration
- Limited flexibility compared to scripting
- Dependency on platform features

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9. CONCLUSION

This project successfully demonstrates how UI Policies can be used to enforce mandatory fields and improve data quality. The use of Update Sets ensures smooth and reliable migration of changes across environments. Overall, the solution enhances system efficiency, reduces errors, and improves user experience.

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10. FUTURE SCOPE

- Integration with advanced workflows
- Automation using scripts and business rules
- Enhanced validation techniques
- Real-time notifications and alerts
- Integration with external systems

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11. APPENDIX

- GitHub Repository Link
- Project Demo Link

Github link

https://github.com/deborashalini123-prog/Enforcing-Mandatory-Fields-Using-UI-Policies-and-Migrating-Changes-with-Update-Sets.git

Project demo link

https://drive.google.com/file/d/1bKOj_HuvEPW6QVkiv1T5wHP3f-r_9zVs/view?usp=drivesdk